

Hospital Admissions Data Analysis

Summary Report

Power BI Dashboard Analysis

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Project: Power BI Internship Project

Institution: Tech Studio Academy

Year: 2026

Turning hospital admission data into actionable insights through data analysis and visualization.

1. Executive Summary

This project analyzes hospital admission data using MySQL and Power BI to identify trends in patient admissions, readmissions, treatment costs, and patient outcomes. The data was prepared and analyzed before being presented through an interactive dashboard designed to support informed healthcare decision-making.

Key findings include a high 30-day readmission rate in Cardiology, increased readmission risk among patients aged 61 and above, October as the month with the highest readmission rate, and ICU as the highest-cost department. The analysis also indicates that higher treatment costs do not always result in better patient outcomes.

2. Project Overview

The purpose of the project was to transform hospital admission records into meaningful performance insights. The analysis focused on patient admissions, hospital performance, high-risk patient groups, treatment costs, and patient outcomes.

Project Objectives

- Analyze patient admissions across hospital departments.
- Monitor hospital performance using key performance indicators.
- Identify patient groups with higher readmission risk.
- Evaluate treatment costs across departments and outcomes.
- Track and compare patient outcomes.

3. Dataset Overview

The dataset contains patient admission records collected from different hospital departments. The analysis used 1,000 patient records across eight departments and covered January to December.

Dataset Item	Description
Total Patients / Records	1,000
Departments	8
Age Groups	0–18, 19–40, 41–60, 61+
Patient Outcomes	Recovered, Improved, Referred, Deceased
Period Covered	January – December

4. Data Cleaning and Preparation

Before analysis, the dataset was prepared in Power BI to improve consistency and ensure that the fields could be used correctly in the visualizations and calculations.

The preparation process included:

- Importing the CSV dataset into Power BI.
- Checking for missing values.
- Removing duplicates.
- Correcting data types.
- Creating an Age Group column using Power Query.
- Building DAX measures for the key performance indicators.

5. Data Analysis and DAX Measures

DAX measures were created to calculate the core hospital performance metrics used in the dashboard.

Measure	Purpose
Total Patients	Counts the patient admission records.
Average Length of Stay	Calculates the average number of days patients stayed in hospital.
Average Treatment Cost	Calculates the average treatment cost per patient.
30-Day Readmission Rate	Calculates the proportion of patients readmitted within 30 days.

6. Dashboard Development

The Power BI dashboard was designed to provide a clear overview of hospital performance and allow users to explore the data interactively. The required analysis was organized around hospital overview, readmission analysis, and cost and outcome analysis.

Hospital Overview

The overview presents the main KPIs, patient admissions by department, and interactive Department and Discharge Month slicers. The reported KPI values include 1,000 total patients, an average length of stay of 7.46 days, an average treatment cost of approximately 330.28K, and an overall 30-day readmission rate of 21%.

Readmission Analysis

The readmission analysis examines monthly trends, readmission rates by department and age group, and departmental performance using average length of stay and readmission rate.

Cost and Outcome Analysis

The cost and outcome analysis examines patient outcome distribution, average treatment cost by outcome, the relationship between length of stay and treatment cost, and departmental cost and readmission performance.

7. Key Findings

Patient Admissions

The hospital recorded 1,000 patient admissions. Emergency recorded the highest number of admissions, while Oncology recorded the lowest according to the dashboard analysis.

Readmissions

October recorded the highest monthly 30-day readmission rate. Cardiology had the highest departmental readmission rate at 28%, indicating an area that may require closer monitoring.

Patient Risk

Patients aged 61 years and above experienced the highest readmission rate, identifying this group as the most at-risk age category in the analysis.

Treatment Costs

ICU recorded the highest treatment cost, while Oncology was also identified as a high-cost department. The analysis suggests that higher treatment costs do not necessarily translate into better patient outcomes.

Patient Outcomes

Recovered patients represented the largest outcome category in the dashboard analysis.

8. Business Insights

The findings provide several areas for healthcare management attention. High readmission rates in Cardiology and among patients aged 61+ suggest opportunities for stronger discharge planning and follow-up care. High treatment costs in departments such as ICU and Oncology also highlight the need for continued monitoring of resource use and patient outcomes.

9. Recommendations

- Improve discharge planning for elderly patients.
- Strengthen post-discharge follow-up to help reduce avoidable readmissions.
- Review treatment and discharge processes in Cardiology.
- Monitor spending and outcomes in high-cost departments such as ICU and Oncology.
- Continue using Power BI dashboards to monitor hospital performance and support data-driven decisions.

10. Conclusion

The project successfully transformed hospital admission data into an interactive Power BI dashboard that provides insights into patient admissions, readmissions, treatment costs, and patient outcomes. The analysis demonstrates how data visualization and KPI monitoring can help identify high-risk groups, departments requiring attention, and areas for improved resource allocation and patient care.

11. Tools Used

- MySQL – data analysis and preparation.
- Microsoft Power BI – dashboard development and visualization.
- Power Query – data cleaning and transformation.
- DAX – KPI and analytical measure calculations.

End of Report